Mesopore-enhanced charge trapping in TENGs

"What is the mechanism behind mesopore-enhanced charge trapping in TENGs?" *Generated by LeapSpace*

Mesopores enhance charge trapping in TENGs by **creating more internal surface, interface, and pore-wall sites where triboelectric charges can be captured and held**, while also strengthening **interfacial polarization / microcapacitor-like charge confinement** that slows charge decay [48] [90] [100] [101] [101].

Mechanism

Mesopore feature	What it does to charges	Why that helps TENGs
High mesopore surface area	Provides more sites for charge storage and trapping [48] [101]	More trapped charge raises surface potential and transferred charge density [101]
Pore-wall / interface abundance	Charges accumulate at pore boundaries and heterogeneous interfaces [100] [101]	Bound charges are harder to dissipate, so retention improves [101] [101]
Space-charge / interfacial polarization	Local charge imbalance builds dipole-like polarization at interfaces [90] [100] [101]	Polarization helps confine charges and suppress recombination [90]
Confinement in structured pores	Mesoporous networks act like internal charge reservoirs or microcapacitors [90] [101]	Stored charges persist longer, supporting higher and more stable output [90] [101]

The core idea

Mesopores do **not** just add empty space; they create a **hierarchical trap landscape**. In the retrieved evidence, the best-supported explanation is that mesoporous structures increase **accessible surface area**, which increases the number of **trap sites**, and they also create **heterogeneous interfaces** where Maxwell-Wagner-type polarization and charge accumulation occur [48] [90] [100] [101]. That combination reduces charge diffusion and decay, so more triboelectric charge remains available for electrostatic induction [101] [101].

Bottom line

Mesopore-enhanced charge trapping in TENGs works mainly through surface-area-driven trap multiplication plus interface-driven charge confinement. The mesoporous network gives charges more places to land, while the pore boundaries and heterogeneous interfaces keep those charges localized long enough to improve retention, transferred charge density, and output stability [48] [90] [101] [101].

Confidence level: **High** — multiple retrieved TENG sources independently support the same trapping-by-surface-area-and-interface-polarization mechanism, including direct evidence for mesoporous SiO_2 charge-storage sites and hierarchical charge-confinement layers [48] [90] [100] [101] [101].

References

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